

1. MACHINE OVERVIEW

The Press-2000 is a 2000-ton hydraulic shop press used for stamping and forming metal sheets and components. It operates on a dual-pump hydraulic system rated for ISO-VG68 hydraulic fluid at 210 bar maximum operating pressure. This manual covers serial prefixes HY- and HP- only.

WARNING: The Press-2000 hydraulic system stores significant energy even when the pump is off. Always follow the pressure bleed procedure (§12.4) before opening any hydraulic line.

The Press-2000 uses a different error code scheme from the RoboInject line. Always confirm the machine model before applying any corrective action. Section 2 of this manual provides a complete cross-reference.

1.1 DAILY OPERATOR CHECKS

- Visually inspect hydraulic fluid level in the reservoir sight glass.
- Check pressure gauge reading at idle (should be 0-5 bar).
- Verify all safety light curtains are clean and unobstructed.
- Listen for unusual pump cavitation sounds during warm-up.

3. ERROR CODE REFERENCE

3.1 E101 ? Low Hydraulic Pressure

IMPORTANT: This code means something different on the Press-2000 than on the RoboInject line. Do not apply RoboInject corrective actions.

Meaning: The main hydraulic system pressure has fallen below 150 bar during the press stroke. The press will stop immediately and hold at current position.

Probable causes:

- Hydraulic fluid level below the minimum mark in the reservoir.
- Worn pump seal allowing internal leakage.
- Relief valve stuck partially open or set below specification.
- External leak in the main pressure line (between pump and manifold).
- Air ingress into the hydraulic system (aerated fluid).

Corrective action:

Step 1. Check the reservoir sight glass ? fluid must be between the MIN and MAX marks. Top up with ISO-VG68 hydraulic fluid if needed (3.5 L capacity).

Step 2. Inspect the main pump seal (located behind the pump cover on the motor bell housing). Oil drip indicates a failed seal. Replace seal kit P/N PS-2000-SEAL.

Step 3. Test the relief valve: slowly increase pressure in manual mode while watching the gauge. The valve should crack open at 210 bar ± 5 bar. Adjust using the hex nut on the valve body (clockwise to increase).

Step 4. Check all unions and connectors on the main pressure line from pump to manifold. Tighten any loose fittings. Look for wet spots indicating a leak.

Step 5. Bleed air from the system: open the bleed valve at the top of the manifold and run the pump for 30 seconds. Close when steady oil flow appears.

Source: Section 3.1, page 92.

3.2 E204 ? High Pressure Exceeded

Meaning: The hydraulic pressure has exceeded 225 bar (the overpressure limit). The press has triggered an emergency stop.

Probable causes:

- Intensifier valve stuck in the closed position
- Pressure transducer sensor fault
- Control board pressure limit parameter set too high

Corrective action:

Step 1. Reset the emergency stop and clear the fault code.

Step 2. Test the pressure transducer: voltage at pins 1-3 should read 0.5V at idle and 4.5V at 210 bar. Replace if outside this range.

Step 3. In the controller menu, verify parameter PR-LIMIT is set to 210.

Source: Section 5.2, page 138.

4. HYDRAULIC SYSTEM MAINTENANCE

4.1 Pump Service Intervals

The main hydraulic pump (Vickers-style vane pump, P/N VP-2000) should be serviced every 2000 operating hours:

- Replace vane cartridge kit (P/N VC-2000-KIT)
- Replace inlet filter (P/N IF-2000)
- Replace oil return line filter (P/N RF-2000)
- Inspect coupling between pump and motor for wear

4.2 Hydraulic Fluid Specification

Use only ISO-VG68 anti-wear hydraulic fluid. Do not mix with other grades. The system holds approximately 12 liters including lines and cylinder.

Fluid temperature should be 40-55°C during normal operation. If temperature exceeds 65°C, check the oil cooler for blocked fins.

4.3 Pressure Test Procedure

Connect the digital pressure tester to test port TP-1 on the manifold. Run the press through one complete cycle in manual mode. Record peak pressure ? should reach 210 bar $\pm$ 5 bar. If peak is below 190 bar, suspect pump wear or relief valve set too low.

5. ELECTRICAL & SAFETY SYSTEMS

5.1 Controller Fuse Locations

Main control board F1: 10A (PLC power supply)

Main control board F2: 6.3A (24V DC sensor bus)

Pump motor contactor coil: 2A

Hydraulic valve solenoid bank: 4A total

5.2 Light Curtain Alignment

The front and rear safety light curtains (Type 4, P/N LC-2000) must be aligned within $\pm 5^\circ$ of the receiver axis. The emitter LED should be steady green when aligned. Flashing red indicates misalignment or a broken emitter unit.

5.3 Emergency Stop Circuit

E-stop pushbuttons are located at all four corners of the press base. The E-stop circuit is dual-channel, monitored by the safety PLC. If any E-stop is pressed or the circuit wiring is broken between the safety relay and the contactor, the main pump contactor drops out immediately.

To test: press each E-stop individually and verify the controller shows 'E-Stop Channel A/B Open' on the diagnostics screen.